

List

```
movies = ["Inception", "Avatar", "Titanic", "Interstellar", "Gladiator"]
print(movies[0], movies[-1])
```

Inception Gladiator

```
movies.append("Joker")
movies.pop(1)
print(movies)
```

['Inception', 'Titanic', 'Interstellar', 'Gladiator', 'Joker']

```
movies.sort()
print(movies)
```

['Gladiator', 'Inception', 'Interstellar', 'Joker', 'Titanic']

```
print(len(movies))
print("Avatar" in movies)
```

5
False

Tuple

```
cities = ("Chennai", "Mumbai", "Delhi", "Kolkata", "Bangalore")
print(cities[1], cities[3])
```

Mumbai Kolkata

```
c1, c2, c3, c4, c5 = cities
print(c1, c2, c3, c4, c5)
```

Chennai Mumbai Delhi Kolkata Bangalore

```
city_list = list(cities)
city_list.append("Hyderabad")
cities = tuple(city_list)
print(cities)
```

('Chennai', 'Mumbai', 'Delhi', 'Kolkata', 'Bangalore', 'Hyderabad')

Dictionary

```
student = {
    "name": "Sowmiya",
    "age": 20,
    "grade": "A",
    "city": "Trichy"
}
```

```
print(student.keys())
print(student.values())
print(student.items())
```

```
dict_keys(['name', 'age', 'grade', 'city'])
dict_values(['Sowmiya', 20, 'A', 'Trichy'])
dict_items([('name', 'Sowmiya'), ('age', 20), ('grade', 'A'), ('city', 'Trichy')])
```

```
student["grade"] = "A+"
student["email"] = "sowmiya@email.com"
```

```
del student["city"]
print(student)
```

{'name': 'Sowmiya', 'age': 20, 'grade': 'A+', 'email': 'sowmiya@email.com'}

Set

```
weekday = {"Math", "Science", "English"}  
weekend = {"Art", "Music", "Math"}
```

```
print(weekday | weekend)  
print(weekday & weekend)  
print(weekday - weekend)
```

```
{'Music', 'Math', 'Art', 'Science', 'English'}  
{'Math'}  
{'Science', 'English'}
```

```
weekday.add("History")  
weekend.remove("Music")
```

```
print("Math" in weekday or "Math" in weekend)
```

```
True
```

Function

```
def greet(name):  
    print(f"Hello, {name}! Welcome to Day 2.")  
  
greet("Sowmiya")
```

```
Hello, Sowmiya! Welcome to Day 2.
```

```
def add_numbers(a, b):  
    return a + b
```

```
def find_max(a, b, c):  
    return max(a, b, c)
```

```
def is_even(n):  
    return n % 2 == 0
```

Default Parameters

```
def introduce(name, language="Python"):  
    print(f"{name} is learning {language}.")
```

```
introduce("Sowmiya")  
introduce("Sowmiya", "SQL")
```

```
Sowmiya is learning Python.  
Sowmiya is learning SQL.
```

```
def calculate_discount(price, discount=10):  
    return price - (price * discount / 100)
```

```
print(calculate_discount(1000))
```

```
900.0
```

if/else

```
num = int(input("Enter number: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Enter number: 5
Positive

```
score = int(input("Enter score: "))
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
elif score >= 60:
    print("D")
else:
    print("F")
```

Enter score: 95
A

```
year = int(input("Enter year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Enter year: 35
Not a Leap Year

For Loop

```
for i in range(1, 21):
    print(i)
```

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
for i in range(2, 51, 2):
    print(i)
```

2
4
6
8
10
12
14
16
18
20
22
24
26
28

```
30
32
34
36
38
40
42
44
46
48
50
```

```
fruits = ["apple", "banana", "cherry"]
for i, fruit in enumerate(fruits):
    print(i, fruit)
```

```
0 apple
1 banana
2 cherry
```

```
print(sum(range(1, 101)))
```

```
5050
```

```
text = "hello world"
vowels = "aeiou"
count = sum(1 for char in text if char in vowels)
print(count)
```

```
3
```

NumPy Tasks

```
import numpy as np
```

1D Arrays

```
print(np.array([10,20,30,40,50]))
print(np.zeros(10))
print(np.ones(10))
print(np.arange(1,21))
print(np.linspace(0,1,7))
print(np.random.randint(1,101,8))
```

```
[10 20 30 40 50]
[0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
[1. 1. 1. 1. 1. 1. 1. 1. 1. 1.]
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
[0.          0.16666667 0.33333333 0.5          0.66666667 0.83333333
 1.          ]
[18 88 50 22 93 73 80 92]
```

2D Arrays

```
print(np.zeros((3,3)))
print(np.eye(3))
print(np.full((4,4),7))
print(np.random.rand(3,4))
print(np.array([[1,2,3],[4,5,6],[7,8,9]]))
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[[7 7 7]
 [7 7 7]
 [7 7 7]
 [7 7 7]]
[[0.75627061 0.98977944 0.64648873 0.1595041 ]
 [0.94242341 0.19982873 0.46996194 0.70865822]
 [0.3948213  0.71038298 0.80126682 0.03869774]]
[[1 2 3]
```

```
[4 5 6]
[7 8 9]]
```

Attribute

```
arr = np.random.rand(4,5)

print(arr.ndim)
print(arr.shape)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
2
(4, 5)
20
float64
8
160
```

Data Types

```
a = np.array([1,2,3])
print(a.dtype)

b = np.array([1.1,2.2,3.3])
print(b.dtype)

print(a.astype(float))

c = np.array([1,2.5,3])
print(c.dtype)

d = np.array(["a","b","c"], dtype=str)
print(d.dtype)
```

```
int64
float64
[1. 2. 3.]
float64
<U1
```

Special Arrays

```
arr = np.zeros(10)
arr[5] = 99
print(arr)

print(np.arange(0,51,5))

eye = np.eye(5)
print(np.diag(eye))

a = np.array([1,2,3])
b = np.array([4,5,6])
print(np.vstack((a,b)))
print(np.hstack((a,b)))
```

```
[ 0.  0.  0.  0.  0. 99.  0.  0.  0.  0.]
[ 0  5 10 15 20 25 30 35 40 45 50]
[1. 1. 1. 1. 1.]
[[1 2 3]
 [4 5 6]]
[1 2 3 4 5 6]
```

Indexing&Slicing

```
arr = np.arange(1,16)
print(arr[:5])
print(arr[-3:])
print(arr[::-2])
```

```
mat = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(mat[1,2])
print(mat[1,:])
print(mat[:,2])
```

```
[1 2 3 4 5]
[13 14 15]
[ 1  3  5  7  9 11 13 15]
6
[4 5 6]
[3 6 9]
```

Operations

```
a = np.array([1,2,3,4,5])
b = np.array([5,4,3,2,1])

print(a+b, a-b, a*b, a/b)

arr = np.arange(1,11)
print(arr*3)

print(np.sqrt([4,9,16,25,36]))

arr = np.array([1,2,3,4,5])
print(arr.sum(), arr.mean(), arr.min(), arr.max())

mat = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(mat.sum(axis=1)) # row sum
print(mat.sum(axis=0)) # column sum
```

```
[6 6 6 6 6] [-4 -2  0  2  4] [5 8 9 8 5] [0.2 0.5 1.  2.  5. ]
[ 3  6  9 12 15 18 21 24 27 30]
[2.  3.  4.  5.  6.]
15 3.0 1 5
[ 6 15 24]
[12 15 18]
```